SUMMARY SECURITY REPORT · for students and general readers

test_flows.csv

Analysis #1 · report #1 · generated 2026-10-28 16:20 UTC · detector v3 · report schema 2.1

Scope. Offline analysis of 300 completed network flows uploaded as "test_flows.csv". The file contains flow measurements only: no live traffic, IP addresses, ports, timestamps or payloads. The results are predictions of a machine-learning model: they are not confirmed incidents and they do not predict future attacks.

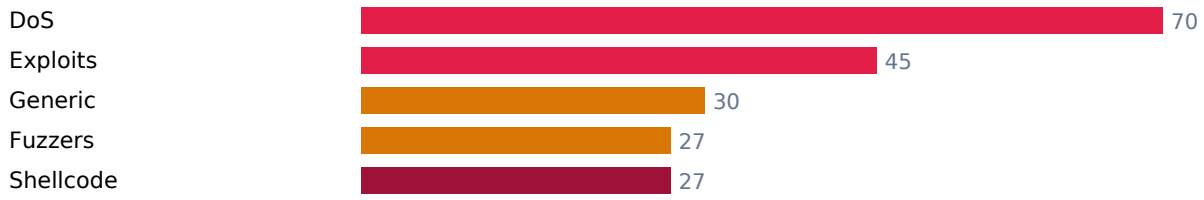
1. Overall flow summary

Flows analysed	Predicted Normal	Predicted attack	Need review (confidence below 70%)
300	**34 (11.3%)**	**266 (88.7%)**	**151**

Review priority: Critical — a fixed rule of this application (rule: serious flows >= 25% of all flows, or malicious flows >= 50%), used to decide how soon to review. It is not a measure of damage and not a confirmed compromise.

How sure is the model?	High (90% or more)	Medium (70-90%)	Low (below 70%)
Attack or normal?	263	26	11
Exact attack type?	92	57	151

The model is usually sure whether a flow is an attack; it is less sure about the exact type, because some attack types look the same in these measurements.



2. What was detected

Does the analysed traffic show signs of an attack?

Yes. The model found signs of attack-like activity in 266 of 300 flows. These are predictions from flow measurements, not confirmed incidents, and they do not show whether an attack will happen in the future.

Predicted group	Flows	What it means
DoS	70	Traffic resembling attempts to make a service unavailable. Public guidance (a possible explanation, not proof): “Adversaries may perform Network Denial of Service (DoS) attacks to degrade or block the availability of targeted resources to users.” [S1]
Exploits	45	Traffic resembling attempts to use a known weakness in software. Public guidance (a possible explanation, not proof): “Adversaries may attempt to exploit a weakness in an Internet-facing host or system to initially access a network.” [S11]
Generic	30	Traffic resembling attacks that work against block ciphers in general. Public guidance (a possible explanation, not proof): “Adversaries may compromise a network device’s encryption capability in order to bypass encryption that would otherwise protect data communications.” [S19]

Group descriptions: UNSW-NB15 category descriptions (Moustafa & Slay, 2015). Quoted guidance describes this kind of activity in general; it is a possible explanation, not proof that it happened here.

3. Why the traffic was flagged

- 266 of 300 flows (88.7%) were predicted as attacks and 34 as Normal. The largest predicted groups are DoS (70), Exploits (45) and Generic (30).
- For the 70 DoS predictions, the most frequent reason was the average packet size sent by the source (a top-three reason in 62 of 70 flows).
- For the 45 Exploits predictions, the most frequent reason was the number of recent connections to the same service and destination (a top-three reason in 37 of 45 flows).
- For 263 of 300 flows the model is at least 90% sure whether it is an attack; the uncertainty is mostly about the exact type (151 predictions have type confidence below 70%). For 7 flows, Normal traffic is still a possible answer.
- These are predictions about completed flows, not confirmed incidents or a forecast of future attacks. Check the flagged flows against logs before blocking anything.

How to read this: the model looks at measurements of each flow (sizes, counts, timing, service). The reasons above are the measurements that most often moved the model towards a group; they describe how the model decided, not what caused the traffic.

Examples: the reasoning for one flow of each main type

DoS — example flow #284 (evidence: finding F284 in the Detailed report)

1. **How sure the model is:** The model is 58% confident that this flow is DoS with a probability of 0.99 that it is an attack of any type; its 90% prediction set contains Analysis, Backdoor, DoS, and Reconnaissance. (AI, checked)
2. **Main reasons:** The strongest evidence is mean source packet size = 100 and bytes from source = 200, while recent connections: same service and source = 11 argues against DoS. (AI, checked)
3. **How typical the values are:** The value mean source packet size = 100: at least half of training DoS flows have exactly this value; the middle half of training Normal flows is 59 to 130.
4. **Why not Exploits:** Compared with Exploits, the second most likely class (15.9%), column mean source packet size = 100 favours DoS by +0.467 in model score. (AI, checked)
5. **In this file:** In the analysed dataset, 70 flows (26.3% of the predicted attacks) were classified as DoS, with average confidence 0.56. (AI, checked)

Exploits — example flow #134 (evidence: finding F134 in the Detailed report)

1. **How sure the model is:** The model gives Exploits a probability of 0.66 and is 0.83 sure this flow is an attack of any type; its 90% shortlist is Exploits.
2. **Main reasons:** The strongest evidence is recent connections: same source and destination = 1 and source time-to-live (TTL) = 62. (AI, checked)
3. **How typical the values are:** The value source time-to-live (TTL) = 62 occurs in 40.0% of training Exploits flows and 3.9% of training Normal flows. (AI, checked)
4. **Why not Normal:** Compared with Normal, the second most likely class (17.4%), column source time-to-live (TTL) = 62 favours Exploits by +2.456 in model score. (AI, checked)
5. **In this file:** In the analysed dataset, 45 flows (16.9% of the predicted attacks) were classified as Exploits, with average confidence 0.67. (AI, checked)

Points marked (AI, checked) were written by the local language model (qwen2.5:7b) and every number and direction was checked against the evidence; the other points are written by code from the same evidence. Typical values come from the training flows only. Each example explains one flow, not every flow of the type.

4. What to do

Verify now

What to do	How to do it	When it applies	How to check success
Review the low-confidence predictions A1 · Security analyst	Open the flow list filtered to confidence Low and review the 151 flows below 70% before acting on any of them.	Always; read-only.	Each reviewed flow is marked confirmed, false positive or unclear in the checklist.
Check flow #284 in logs or captures A2 · Security analyst	Find flow #284 (unas/-, state INT, 200 bytes sent, 0 received) in packet captures or sensor logs from the same period and decide whether it is DoS activity. [S1] [S31]	Always first; read-only. Nothing should be blocked on the prediction alone.	Confirmed if logs or captures show activity matching the cited description of DoS; weakened if the traffic belongs to a known service or job of your own team, or other sensors show nothing for that period.

Contain only if confirmed

What to do	How to do it	When it applies	How to check success
Filter or block the confirmed traffic A7 · Network engineering	Filter Network Traffic: When flood volumes exceed the capacity of the network connection being targeted, it is typically necessary to intercept the incoming traffic upstream to filter out the attack traffic from ... [S7]	Packet captures or sensor logs confirm the traffic is malicious AND its source, destination and service are identified (the uploaded file has none of these). A DoS prediction alone is not enough.	Sensor counts for the matching traffic drop while service health checks stay normal.
Reset exposed credentials or keys A9 · System administrator / identity team	Change system admin passwords, rotate private keys and service/application account secrets where compromise is suspected revoke privileged access. [S18]	Authentication logs or endpoint alerts show that a named account, password or key was exposed or misused. Network-flow predictions cannot show this. A Generic prediction alone is not enough.	The old credentials are rejected and no new suspicious sign-ins appear.

Recover and monitor

What to do	How to do it	When it applies	How to check success
Restore and monitor after a confirmed incident A10 · Incident response lead with the system owners	See NIST SP 800-184: Guide for Cybersecurity Event Recovery for additional guidance. [S32]	Only after a confirmed incident has been contained.	Restored systems behave normally and monitoring shows no recurrence for an agreed period.

This application never changes your network: these are suggestions for a person to decide on. The full list with evidence is in the detailed report.

5. How to check the outcome

Evidence you need. Packet captures, firewall or sensor logs and host logs from the period when the flows were recorded. The uploaded file has no addresses, ports or timestamps, so flows must be matched by protocol, service, state and byte counts.

Confirms the concern. Another source (logs or captures) shows the same activity, for example the same service, volumes and connection pattern as the flagged flows.

Weakens the concern. the flagged traffic belongs to a known service, backup, monitoring or test run by your own team; other sensors or logs from the same period show nothing unusual; Normal is still on the shortlist (7 flows).

Fix worked. After a confirmed problem is fixed, the activity no longer appears in new captures or logs, and normal services still work.

6. Limits and sources

- Predictions are not confirmed incidents; class-to-ATT&CK mappings are hypotheses for context.
- No IP addresses, ports, host names, timestamps or payloads are in the uploaded file, so hosts, timelines and botnet or DNS attribution cannot be established.
- Detailed explanations cover 6 example flows of 300; other flows of the same type may differ.

Sources

- [S1] MITRE ATT&CK, T1498
- [S7] MITRE ATT&CK mitigation, M1037 Filter Network Traffic (for T1498)
- [S11] MITRE ATT&CK, T1190
- [S18] CISA IR Playbooks, page 31
- [S19] MITRE ATT&CK, T1600
- [S31] NIST SP 800-61r3, page 36
- [S32] CISA IR Playbooks, page 15

Same analysis as the detailed report #1 (identical counts, predictions and evidence); the detailed report adds the full evidence, source passages and audit information.